Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Inertial frame

Meaning in my words How would you explain inertial frame to a friend?	**How can you tell?** A puck glides straight at steady speed. Would a camera moving straight at constant speed also see steady motion?
Example or sketch Which bus ride could serve as an inertial frame for watching a puck: cruising straight or turning?	**Non-example and why** A bus brakes and a loose puck seems to surge forward inside it. Does that prove a forward force acted on the puck?

Net force

Meaning in my words How would you explain net force to a friend?	**How can you tell?** Two people push one cart in opposite directions. Why do the arrow directions matter?
Example or sketch One person pushes 6 N right and another 4 N left. What single arrow replaces both?	**Non-example and why** A student adds those pushes to get 10 N right. Which push did they count the wrong way?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Inertia

Meaning in my words What does inertia mean? Say it the way you'd explain it to a friend.	**How can you tell?** An empty cart and a loaded cart both start at rest. We push each with the same net force for one second. Which gains less speed? Why?
Example or sketch A cart is already rolling right. The forces on it balance. Where would you draw it one second later? And one second after that?	**Non-example and why** Someone points to the hand pushing the cart and says, "That push is inertia." What would you tell them?

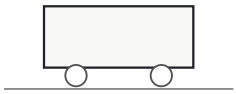
Equilibrium

Meaning in my words How would you explain equilibrium to a friend?	**How can you tell?** A cart rolls straight at steady speed. What must the horizontal forces add up to?
Example or sketch Sketch the horizontal forces on a cart with a 3 N pull right and 3 N friction left.	**Non-example and why** A tossed ball is briefly still at the top. Is it in equilibrium there?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a moving cart with balanced force arrows in orange and its separate velocity arrow in purple.



My marks mean

One camera. Which window?

09

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use timed sightings to test a forecast and explain its limits.

CHOOSE ONE JOB TOGETHER

TRACKING ENGINEER ☐

Radio your camera choice. Use two sightings and a signed velocity to support your call.

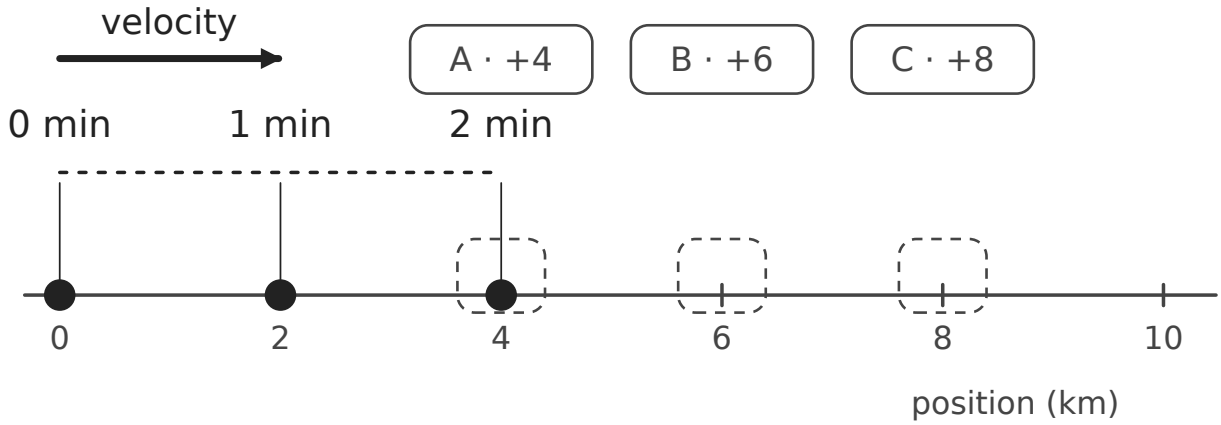
SCIENCE BROADCASTER ☐

Tell viewers where to look. Use your cart drawing to explain why XJ can keep moving without an engine.

Both partners point to a clue. Ask: What has to stay the same for our call to work?

OBSERVATION DESK

One picture at minute 3. Choose A, B or C.



01 OUR FIRST CALL

Where should the camera look? Why?
Write your call. Include a signed velocity and say what you assume.

02 OUR REVISED CALL

Wait for your teacher. Keep your first call.

New time: _____ min Position: _____ km

How far did XJ move in the last minute?
Use that result in your revised call.

On the graphic: add a dashed circle for your forecast and a star for the new sighting. Color the velocity arrow **PURPLE**. Use **PENCIL** for data marks.

Discuss: Why could a straight-line forecast work for a minute even while gravity pulls on XJ? Switch speakers; point to the evidence.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks